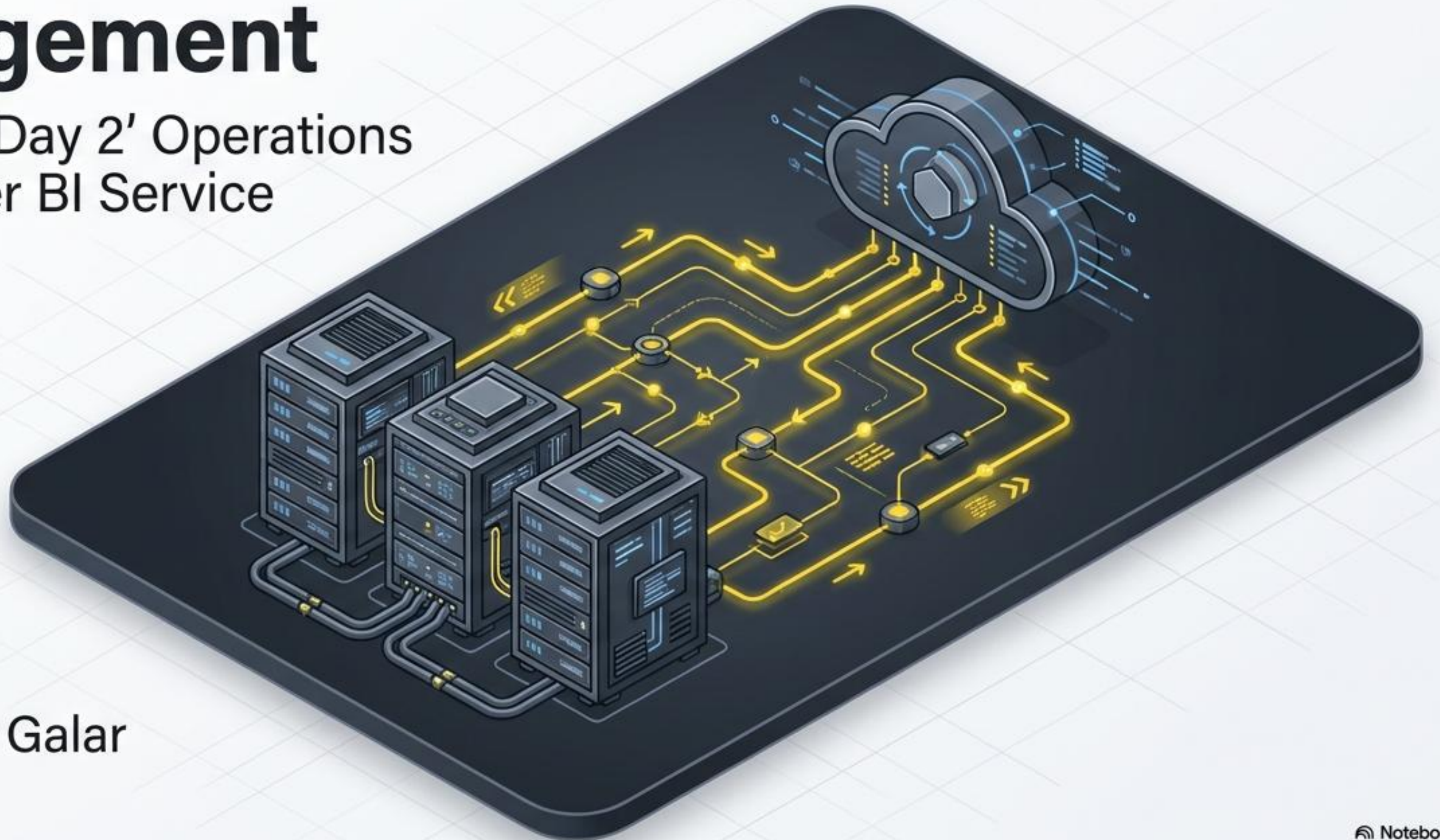


Power BI: Data Sources Management

Mastering 'Day 2' Operations
in the Power BI Service



dr Zbigniew Galar

The Four Pillars of Dataset Deployment

Publishing your report is only the beginning. Successful deployment requires managing four critical operational pillars.



The License & Refresh Matrix

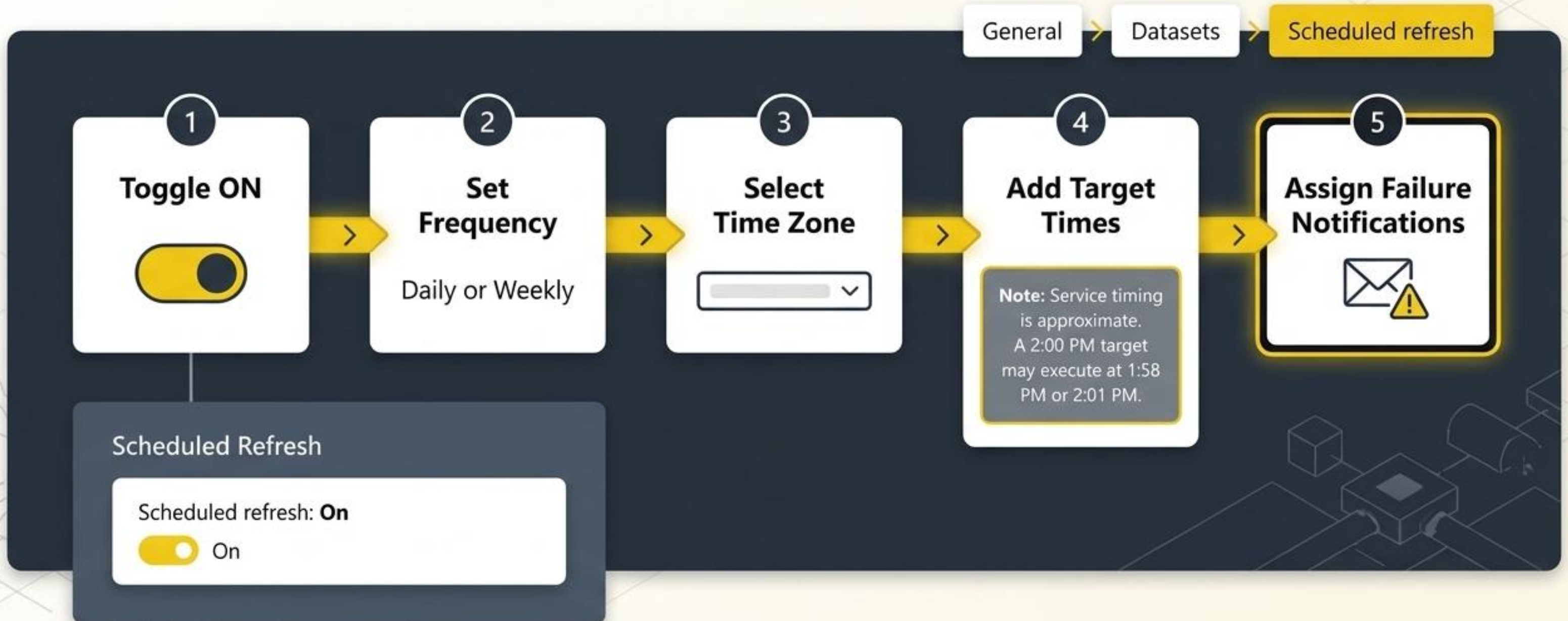
Automated refresh capabilities are strictly governed by your workspace licensing.

Free License	Pro License	Premium
Manual refreshes only. No automation available.	Up to 8 scheduled refreshes per day.	(Per User or Capacity) Up to 48 scheduled refreshes per day.



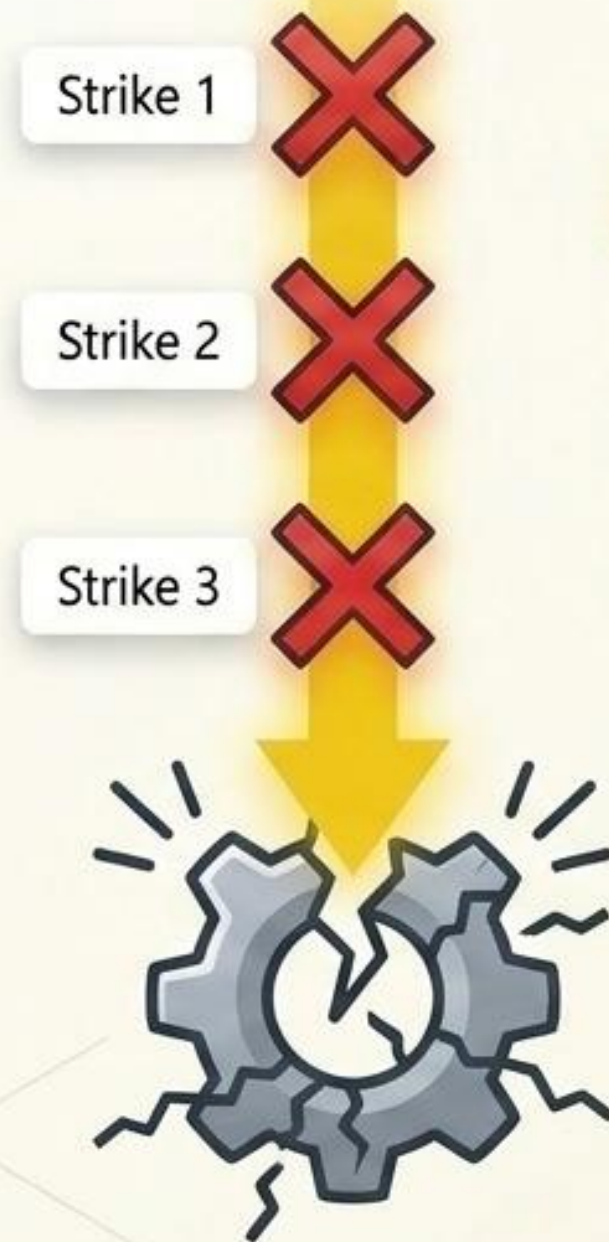
The Refresh Configuration Pipeline

Scheduled refreshes are disabled by default. Follow this exact sequence under **Settings > Datasets**.



The 'Three-Strike' Suspension Rule

Failure to assign a responsive dataset owner in the notification settings can lead to permanent data stagnation.



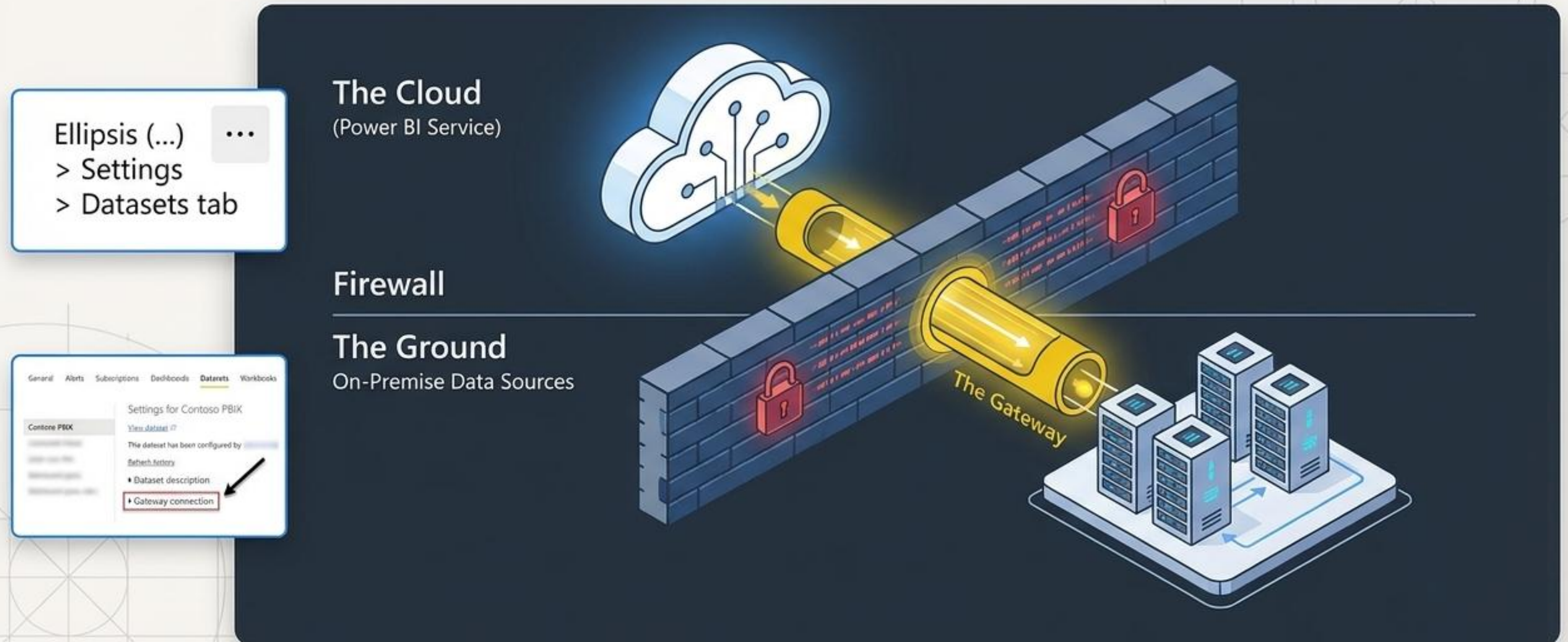
If a scheduled refresh fails three consecutive times, Power BI automatically disables the refresh schedule entirely.



Crucial: This is why Step 5 (Assign Notifications) is mandatory. Without an alert, a disabled schedule goes unnoticed.

The Gateway Bridge

If the Scheduled Refresh menu is grayed out, the Power BI Service cannot reach your data. You need a bridge.



Gateway Infrastructure Selection

Gateways must match the organizational scope and security requirements of the dataset.

Data sources On-premises data gateways Virtual network data gateways

In the Manage Gateways menu, the Data Sources and Data Gateways can be viewed.

Personal Mode

Runs on a local PC using personal credentials.

⚠ Critical Constraint: Only active while the device is powered on and connected to the source.

On-Premise (Standard)

For enterprise **IT**. Allows multiple users to access data behind a firewall without relying on a single user's laptop.

Virtual Network (VNet)

Enterprise-grade cloud and virtual network security.

Gateway connection

Use an On-premises or VNet data gateway

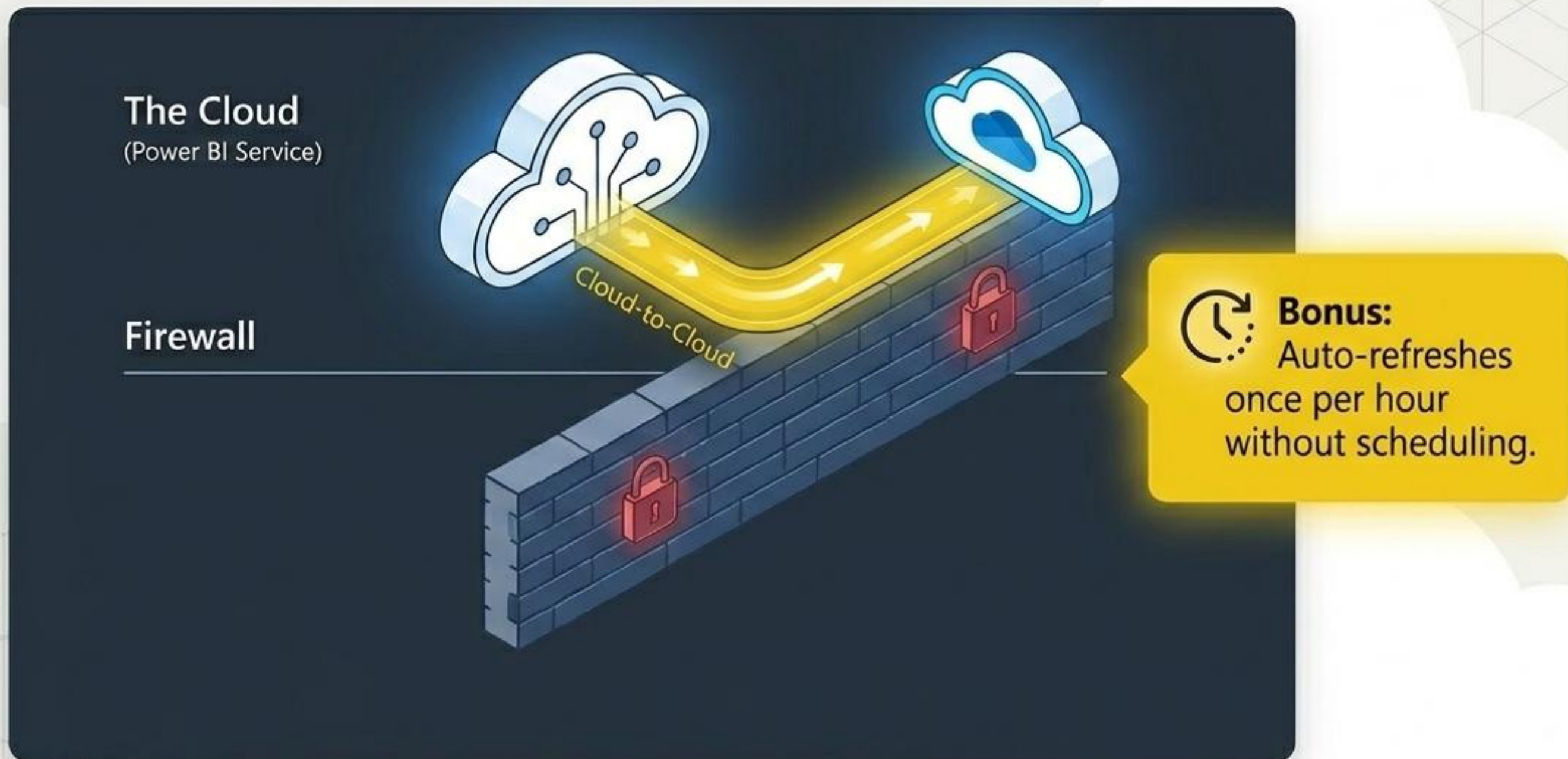
☒ On

Gateway	Department	Contact information	Status	Actions
Personal Gateway			✓ Running on DESKTOP-XXXXXXX	

A Personal Gateway running on a personal computer.

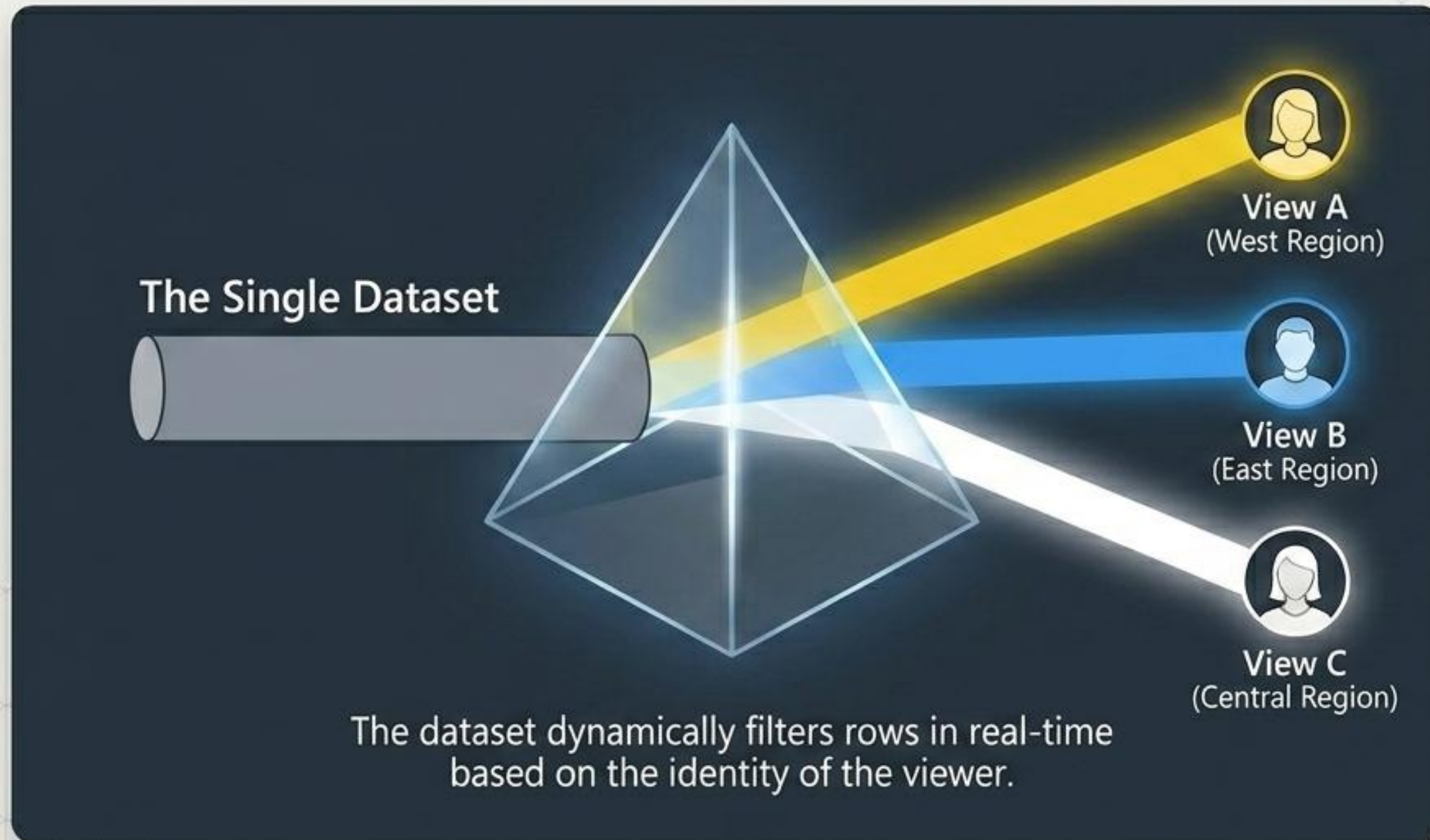
The Cloud Exception: OneDrive & SharePoint

Excel and CSV files hosted in an associated workspace OneDrive do not require a gateway.



The Row-Level Security Prism

Security roles are defined in Power BI Desktop, but user identities must be assigned in the Service.



Assigning Identities to Roles

Map Active Directory individuals or Security Groups to the PBIX roles under **Settings > Security**.

The screenshot illustrates the process of assigning identities to roles in Power BI. On the left, the 'Security' option is highlighted in the left-hand menu. A yellow arrow points from this menu item to the 'Row-Level Security' configuration window. In this window, the role 'Computer Category Manager' is selected. A list of people or groups is shown on the right, with 'Miriam Graham' highlighted. A yellow callout box points to this list with the text: 'The Service connects the **Human** to the **Logic**.'

Row-Level Security

Role

Computer Category Manager (0)

People or groups who belong to this role

Search

Miriam Graham
MiriamG@altentertraining.onmicrosoft.com

Henrietta Mueller
Henrietta@altentertraining.onmicrosoft.com

Megan Bowen
Megan-Bow@onmicrosoft.com

Erwan Mons
Erwan.Mons@onmicrosoft.com

irvara
irvara@onmicrosoft.com

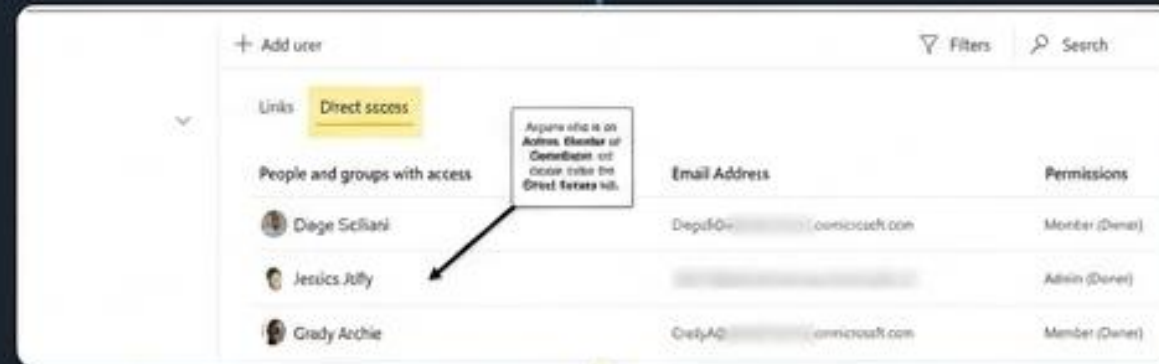
The Service connects the **Human** to the **Logic**.

The Governance Hierarchy & "Security Sprawl"

Managing who can share and who can build from your data.

Workspace Members (Admins, Members, Contributors)

Inherent Access: **SHARE + BUILD**



The screenshot shows a 'Direct access' tab in a workspace sharing interface. It lists three users with their email addresses and permissions. A callout box points to the first user, stating: 'Anyone who is an Admin, Member or Contributor can share the Direct Access link.'

People and groups with access	Email Address	Permissions
Dege Sciani	Dege.D@com.microsoft.com	Member (Owner)
Jessica Jolly		Admin (Owner)
Grady Archie	Grady.A@com.microsoft.com	Member (Owner)

Unchecked sharing creates Security Sprawl

External Users / Non-Members



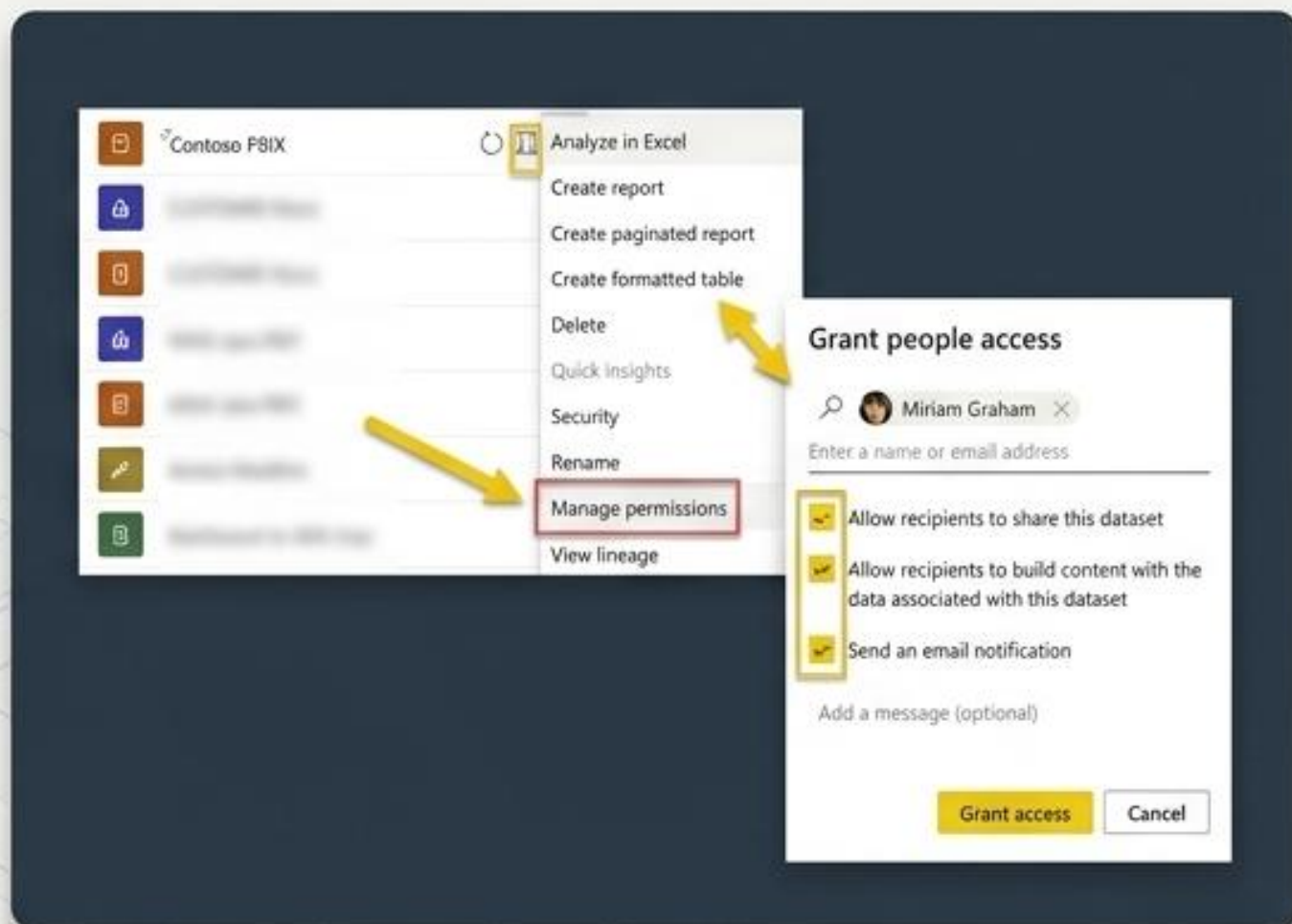
Warning: Sharing with non-workspace members requires manual **boundary controls** to prevent unauthorized downstream reporting.

Direct Access vs. Link Generation

Two distinct paths for provisioning access, requiring interaction with different assets.

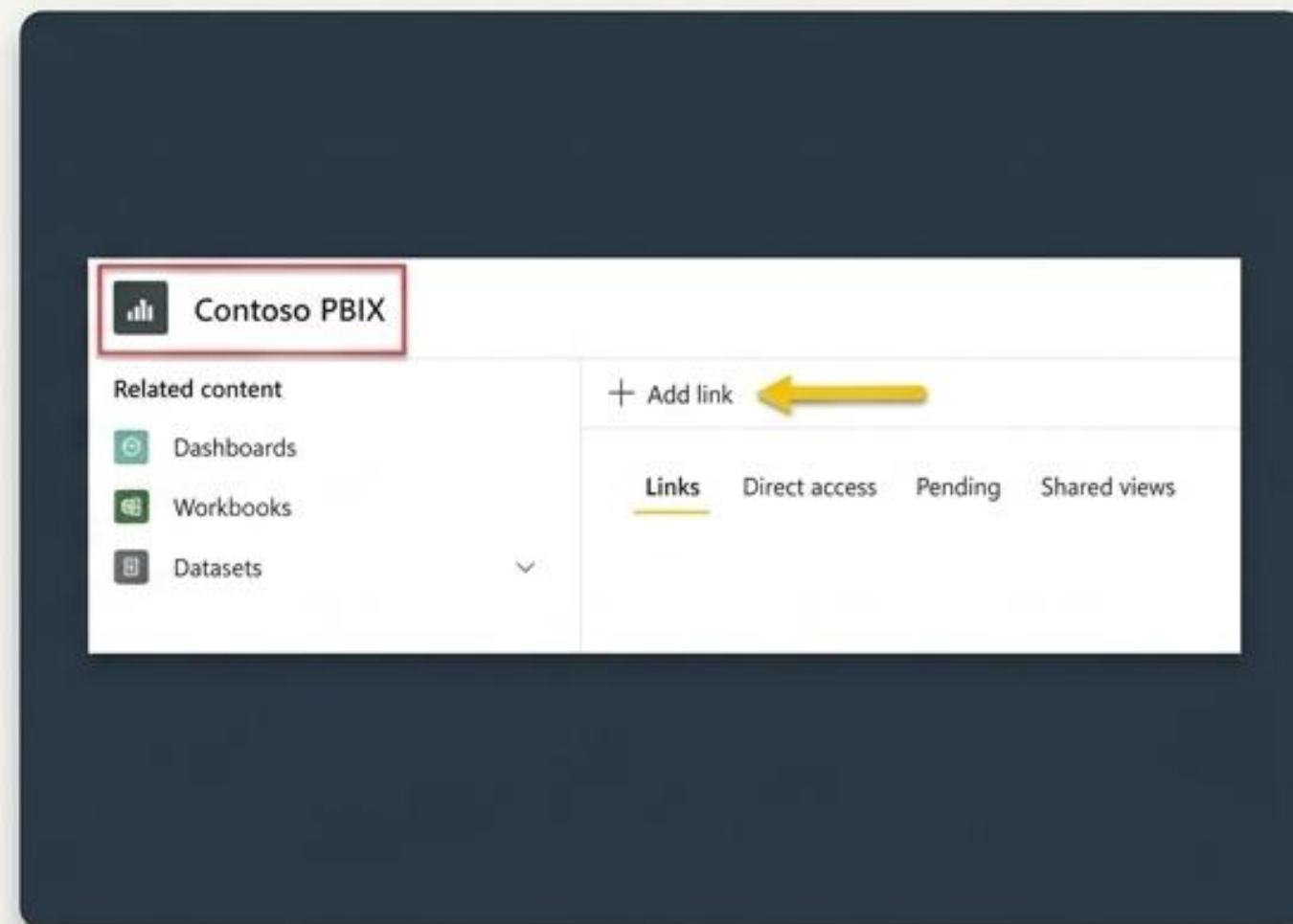
Path A: Direct Access

Granted at the DATASET level.
Used for explicit individuals.



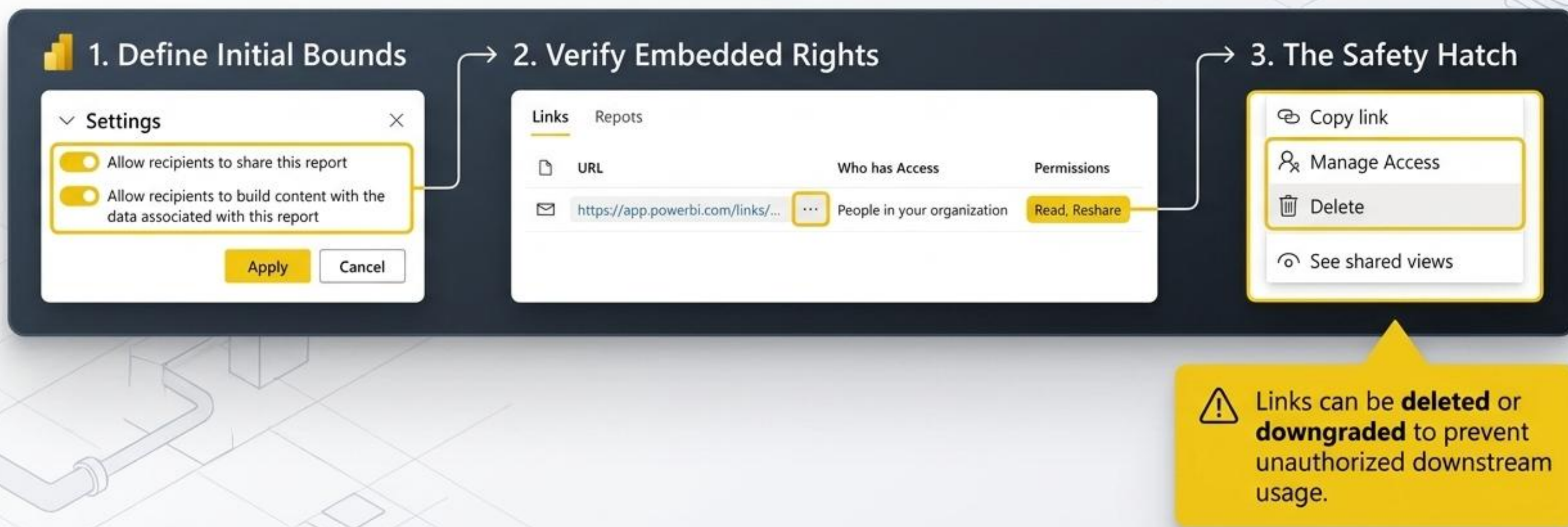
Path B: Shared Links

Granted at the REPORT level.
Used for broader organizational sharing.



Enforcing Link Boundaries

Generated links carry embedded permissions that can be modified or revoked at any time.



The Domino Effect of Dataset Management

These settings do not exist in isolation. A failure in infrastructure directly impacts governance and user trust.



Mastery requires aligning all four pillars simultaneously.